

Course Review

Econ 502: Advanced Microeconomics

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Roadmap

What We Covered

- **Part I - Competitive Markets** (Lectures 1–5): consumer theory, production, partial equilibrium, general equilibrium, welfare theorems
- **Part II - Market Power** (Lectures 6–7): monopoly, price discrimination, oligopoly (Cournot, Bertrand, Stackelberg)
- **Part III - Market Failures** (Lectures 8–9): adverse selection, externalities, public goods
- **Part IV - Game Theory** (Lecture 10): Nash equilibrium, mixed strategies, sequential games, repeated games

How to Use This Review

- For each part: **key tools** + **what trips students up** + **one worked problem**
- Cheat-sheet of formulas at the end of each section
- Stop me with questions whenever - that's what this is for

Part I: Competitive Markets

Consumer Theory: The Toolkit (Lec 1–2)

- **Preferences and utility:** axioms (completeness, transitivity, monotonicity, convexity); MRS = ratio of marginal utilities = slope of indifference curve
- **Utility maximization (primal):** $\max U(x, y)$ s.t. $p_x x + p_y y = I$.
Tangency: $MRS = p_x/p_y$. Yields **Marshallian demand** $x^M(p, I)$ and **indirect utility** $V(p, I)$
- **Expenditure minimization (dual):** $\min p_x x + p_y y$ s.t. $U(x, y) = \bar{u}$. Yields **Hicksian demand** $x^h(p, \bar{u})$ and **expenditure function** $e(p, \bar{u})$
- **Duality:** V and e are inverses in income/utility

Welfare Measures (Lec 2)

For a price change $p_0 \rightarrow p_1$:

- **Compensating Variation:** $CV = e(p_1, u_0) - e(p_0, u_0)$ - extra income at *new* prices to keep old utility
- **Equivalent Variation:** $EV = e(p_1, u_1) - e(p_0, u_1)$ - income change at *old* prices that replicates new utility
- **Consumer surplus:** area below Marshallian demand above price; lies between CV and EV for normal goods

Slutsky decomposition:
$$\frac{\partial x^M}{\partial p} = \underbrace{\frac{\partial x^h}{\partial p}}_{<0 \text{ subst.}} - x \cdot \underbrace{\frac{\partial x^M}{\partial I}}_{\text{income}}$$

Things to Watch For

- **MRS sign convention:** absolute value vs. signed; be consistent
- *CV* vs *EV* confusion: *CV* uses *original* utility; *EV* uses *new* utility
- Cobb-Douglas shortcut: $x^M = \alpha I / p_x$ (only for $U = x^\alpha y^{1-\alpha}$); don't apply to other utilities
- Income vs. substitution effect: substitution effect always against the price change for normal goods

Practice Problem: A Sugary-Drink Tax

A consumer has preferences $U(s, x) = s^{0.3} x^{0.7}$ over sugary drinks s and a composite good x . Income is $I = \$300$ per month. Initially $p_s = \$0.05$ per ounce, $p_x = \$1$. The city imposes a tax $t = \$0.015$ per ounce, raising the consumer price to $p'_s = \$0.065$.

- Find the optimal bundle and utility before and after the tax.
- Compute the **compensating variation** for the price change.
- Compute tax revenue $R = t \cdot s^{**}$ and deadweight loss $DWL = CV - R$.
- If a rebate returns all revenue as a lump sum, is the consumer back to where she started? Why or why not?

Production and Cost (Lec 3)

- **Production function** $Q = f(K, L)$: marginal product, MRTS, isoquants
- **Returns to scale**: if $f(\lambda K, \lambda L) = \lambda^k f(K, L)$, RTS = k (decreasing if < 1 , constant if $= 1$, increasing if > 1)
- **Cost minimization**: tangency $MRTS = w/r$ + isoquant constraint
- **Cost function**: $C(Q, w, r)$. Average cost AC , marginal cost MC . Note: MC crosses AC at minimum

Note: RTS is about scaling *all* inputs; DRS in production $\neq$ DRS in $C(Q)$ (cost function curvature is opposite).

Market Equilibrium and Tax Incidence

(Lec 4)

- **Equilibrium:** $Q^D(P) = Q^S(P)$. Surplus = areas under demand and above supply
- **Per-unit tax** t on producers: solve $Q^D(P^D) = Q^S(P^D - t)$
- **Incidence formula:** consumer share = $\frac{\epsilon_S}{|\epsilon_D| + \epsilon_S}$, producer share
$$= \frac{|\epsilon_D|}{|\epsilon_D| + \epsilon_S}$$
- **DWL** = $\frac{1}{2}(\Delta Q)(t)$; equivalently $\frac{1}{2} \cdot \frac{|\epsilon_D| \epsilon_S}{|\epsilon_D| + \epsilon_S} \cdot \frac{t^2}{P^*} \cdot Q^*$

Key insight: statutory incidence (who writes the check) $\neq$ **economic** incidence (who bears the burden). The latter depends entirely on relative elasticities.

General Equilibrium and Welfare Theorems (Lec 5)

- **Edgeworth box:** allocations where indifference curves are tangent (= equal MRS) are Pareto efficient - the **contract curve**
- **First Welfare Theorem:** any competitive (Walrasian) equilibrium with locally non-satiated preferences is Pareto efficient
- **Second Welfare Theorem:** any Pareto efficient allocation can be supported as a competitive equilibrium after appropriate **lump-sum transfers** of endowments (requires convex preferences/technology)

Slogan: *Markets handle efficiency; government handles equity.* Caveat: real lump-sum transfers are hard (require knowing types), and many real-world settings violate FWT assumptions (externalities, public goods, market power, asymmetric info - Parts II–IV).

Part II: Market Power

Monopoly (Lec 6)

- **Profit max:** $MR = MC$
- **MR for linear demand** $P = a - bQ$: $MR = a - 2bQ$ (twice the slope, same intercept)
- **Lerner index:** $L = (P - MC)/P = 1/|\varepsilon_D|$ (inverse-elasticity rule)
- **DWL** of monopoly: triangle between demand and MC over output not produced
- Monopolist always sets P on the **elastic** portion of demand ($|\varepsilon_D| > 1$)

Price discrimination:

- 1st degree: charge each buyer their WTP (full surplus extraction)
- 2nd degree: self-selection menus (e.g., quantity discounts)
- 3rd degree: segment by observable type, charge different prices;
 $L_i = 1/|\varepsilon_i|$ in each segment

Oligopoly (Lec 7) - Cournot

Two firms, inverse demand $P = a - bQ$, $Q = q_1 + q_2$, equal MC c :

- **Best response:** $q_i^*(q_{-i}) = \frac{a - c - q_{-i}}{2b}$
- **Symmetric NE:** $q_i^* = \frac{a - c}{3b}$, $Q^* = \frac{2(a - c)}{3b}$
- **Quantities are strategic substitutes** (BR slopes downward)
- With n firms: $Q^* = \frac{n(a - c)}{(n + 1)b} \rightarrow \frac{a - c}{b}$ as $n \rightarrow \infty$ (competitive limit)

Oligopoly (Lec 7) - Bertrand and Stackelberg

Bertrand (homogeneous): $p_1 = p_2 = c$ (Bertrand paradox - two firms enough for competitive outcome)

Differentiated Bertrand: demand $q_i = a - bp_i + dp_{-i}$. Prices are **strategic complements** (BR slopes up). $p^* > c$ - paradox resolved by differentiation.

Stackelberg: leader moves first. Leader's problem substitutes follower's BR. Result: leader produces *more* than Cournot, follower less. Leader profit *higher*, follower *lower*. ``Top dog'' commitment.

Note: the leader gains from commitment because quantities are strategic substitutes. With strategic complements (prices), being the price leader can *hurt* you.

Practice Problem: Stackelberg

Two firms compete in quantities. Inverse demand $P = 150 - Q$ where $Q = q_1 + q_2$, and both firms have marginal cost $c = 30$.

- Find the **Cournot** Nash quantities, price, and profit per firm.
- Suppose Firm 1 is the **Stackelberg leader**: chooses q_1 first, Firm 2 observes and chooses q_2 . Find the equilibrium quantities, price, and profits.
- Compare: who gains, who loses? Why does first-mover commitment work here?
- Compute consumer surplus and total welfare under each regime. Which is better for consumers?

Part III: Market Failures

Adverse Selection (Lec 8)

- **Lemons problem (Akerlof):** when sellers know quality but buyers don't, the average quality of cars on the market is below the unconditional average $\Rightarrow$ market unravels
- **Insurance:** types differ in risk p_i . Risk-averse u concave $\Rightarrow$ everyone willing to pay above the actuarially fair premium $p_i L$ (the gap is the **risk premium**)
- **Certainty equivalent:** $\sqrt{CE_i} = (1 - p_i)\sqrt{w} + p_i\sqrt{w - L}$;
 $P_i^{\max} = w - CE_i$
- **Pooling premium** $[\lambda p_H + (1 - \lambda)p_L]L$: low-risk types may refuse $\Rightarrow$ unraveling. Government mandate can solve

Key insight: asymmetric information drives a wedge between competitive equilibrium and efficiency - even with rational agents and well-defined property rights.

Externalities and Public Solutions (Lec 9)

- **Externality:** private MC $\neq$ social MC. Unregulated market over-produces (negative ext.) or under-produces (positive)
- **Coase theorem:** with well-defined property rights and zero transaction costs, private bargaining yields efficiency *regardless of who holds the rights* (only distribution differs)
- **Pigouvian tax:** set $t = \$ \text{ marginal damage}$; firms internalize externality
- **Cap and trade:** issue permits = efficient pollution; market trading achieves efficient allocation regardless of initial allocation (Coase logic)

Tax fixes **price**, lets quantity adjust. Cap fixes **quantity**, lets price adjust.
Equivalent under certainty.

Public Goods (Lec 9)

Non-rival + non-excludable goods.

- **Voluntary contribution:** each agent sets their own $MRS_i = \text{\$ price}$ ($= 1$ when public good costs \$1 per unit). Severe under-provision
- **Samuelson condition (efficient):** $\sum_i MRS_i = 1$ - sum of willingness-to-pay equals marginal cost
- **Free-rider problem:** with N symmetric contributors,
 $G^*/G^{**} = 2/(N + 1) \rightarrow 0$
- **Pigouvian-style subsidy:** $s^* = (N - 1)/(N + 1)$ decentralizes efficiency

Practice Problem: Public Good Provision

Three neighbors share a small park. Each has income $Y_i = 60$ and contributes $g_i \geq 0$. Total park quality is $G = g_1 + g_2 + g_3$. Utility is $U_i(x_i, G) = x_i \cdot G$ where $x_i = 60 - g_i$. Each unit of G costs \$1.

- Derive each neighbor's best response $g_i^*(g_{-i})$ and find the symmetric Nash contribution g_i^* and total G^* .
- Find the social optimum G^{**} and equal-private-consumption allocation x_i^{**} .
- Compute G^*/G^{**} . What does this tell us about free-riding?
- Find the subsidy rate s^* on contributions that decentralizes G^{**} .

Part IV: Game Theory

Normal-Form Games (Lec 10)

- **Strict dominance:** strategy s strictly dominates s' if $u(s, s_{-i}) > u(s', s_{-i})$ for all s_{-i}
- **Nash equilibrium:** profile $(s_1^*, \dots, s_n^*)$ such that no one wants to unilaterally deviate. Each plays best response to others
- **Mixed strategies:** when no pure NE exists (e.g., Matching Pennies). Each player **indifferent** in equilibrium between strategies in their support

Note: a Nash strategy doesn't have to be dominant - only a best response to *the equilibrium profile*. Dominance is much stronger.

Sequential Games (Lec 10)

- **Extensive form (game tree):** decision nodes, branches, payoffs at terminals
- **Backward induction:** solve from the end. At each decision node, the player picks her best action given anticipated future play
- **Subgame Perfect Equilibrium (SPE):** Nash in *every subgame*. Rules out non-credible threats
- **Commitment:** moving first can be valuable when actions are strategic substitutes (Stackelberg top dog ' '); hurtful when complements (puppy dog'')

Entry deterrence: sunk capacity makes the threat of post-entry aggression credible. Words alone do not.

Repeated Games (Lec 10)

- **Finitely repeated PD:** unique SPE is defect every period (backward induction unravels cooperation)
- **Infinitely repeated PD with grim trigger:** cooperation sustainable iff $\delta \geq \delta^*$. Critical δ^* from

$$\frac{\pi^C}{1 - \delta} \geq \pi^D + \frac{\delta \pi^N}{1 - \delta}$$

- **Folk theorem:** for δ high enough, almost any individually rational payoff is supportable in SPE

Real-world parallel: ongoing relationships (firms, countries) have no clear last period - closer to infinite horizon. Cooperation depends on patience and the perceived continuation of the relationship.

Practice Problem: Repeated Prisoner's Dilemma

Two firms play the following stage game in each period:

- Show that (Defect, Defect) is the unique Nash equilibrium of the one-shot game.
- Suppose the game is repeated infinitely with discount factor δ . Under **grim trigger**, compute the PV of always cooperating and of deviating once. Find the critical δ^* above which cooperation is sustainable.

Wrap-Up

Cross-Cutting Themes

1. **Optimization with constraints:** consumer chooses (x, y) s.t. budget; firm chooses (K, L) s.t. isoquant; planner chooses allocation s.t. resources. Same Lagrangian machinery
2. **Equilibrium = mutual consistency:** Walrasian (markets clear), Cournot/Bertrand (best responses), Nash (no profitable deviation), SPE (sequential rationality)
3. **Wedges between private and social MC** explain every market failure: externalities, public goods, monopoly, adverse selection
4. **Property rights + incentives** are the toolkit for fixing those wedges (Coase, Pigou, Samuelson, mechanism design)

Final Tips

- Read the problem **twice**; identify which tool is being asked for
- Set up the **Lagrangian / FOC** carefully - most errors come from setup, not algebra
- For oligopoly: write each firm's profit *as a function of its own choice*, then take FOC
- For game theory: draw the tree or matrix; underline best responses
- For welfare measures: state which utility level you're holding fixed (CV vs EV)
- Check your answer for sign and magnitude reasonableness

Good luck!

Office hours and review session details on Canvas. Reach out with specific questions before the exam.